TP1 : Déploiement d'un Portfolio Next.js avec Ansible et Semaphore UI

Etape_1. Installer Ansible

```
arij@arij:~$ sudo apt install -y ansible
Lecture des listes de paquets... Fait
Construction de l'arbre des dépendances... Fait
Lecture des informations d'état... Fait
Les paquets supplémentaires suivants seront installés :
  ansible-core python3-argcomplete python3-dnspython python3-kerberos python3-libcloud python3-lockfile python3-ntlm-auth
  python3-passlib python3-requests-ntlm python3-resolvelib python3-selinux python3-simplejson python3-winrm python3-xmltodict
Paquets suggérés :
  cowsay sshpass python3-trio python3-aioquic python3-h2 python3-httpx python3-httpcore python-lockfile-doc
Les NOUVEAUX paquets suivants seront installés :
  ansible ansible-core python3-argcomplete python3-dnspython python3-kerberos python3-libcloud python3-lockfile python3-ntlm-auth
  python3-passlib python3-requests-ntlm python3-resolvelib python3-selinux python3-simplejson python3-winrm python3-xmltodict
0 mis à jour, 15 nouvellement installés, 0 à enlever et 69 non mis à jour.
Il est nécessaire de prendre 19,5 Mo dans les archives.
Après cette opération, 315 Mo d'espace disque supplémentaires seront utilisés.
Réception de :1 http://tn.archive.ubuntu.com/ubuntu noble/universe amd64 python3-resolvelib all 1.0.1-1 [25,7 kB]
Réception de :2 http://tn.archive.ubuntu.com/ubuntu noble/main amd64 python3-dnspython all 2.6.1-1ubuntu1 [163 kB]
Réception de :3 http://tn.archive.ubuntu.com/ubuntu noble/universe amd64 ansible-core all 2.16.3-0ubuntu2 [1 280 kB]
Réception de :4 http://tn.archive.ubuntu.com/ubuntu noble/universe amd64 ansible all 9.2.0+dfsg-0ubuntu5 [16,4 MB]
Réception de :5 http://tn.archive.ubuntu.com/ubuntu noble-updates/universe amd64 python3-argcomplete all 3.1.4-1ubuntu0.1 [33,8 kB]
Réception de :6 http://tn.archive.ubuntu.com/ubuntu noble/universe amd64 python3-kerberos amd64 1.1.14-3.1build9 [21,2 kB]
```

```
arij@arij:~$ ansible --version
ansible [core 2.16.3]
  config file = None
  configured module search path = ['/home/arij/.ansible/plugins/modules', '/usr/share/ansible/plugins/modules']
  ansible python module location = /usr/lib/python3/dist-packages/ansible
  ansible collection location = /home/arij/.ansible/collections:/usr/share/ansible/collections
  executable location = /usr/bin/ansible
  python version = 3.12.3 (main, Aug 31 2026, 10:18:26) [GCC 13.3.0] (/usr/bin/python3)
  jinja version = 3.1.2
  libyaml = True
```

2. Créer la structure du projet :

```
arij@arij:~$ mkdir -p portfolio-deploy/{inventory,group_vars,roles/{common/tasks,nodejs/tasks,app/{tasks,handlers},nginx/{tasks,handl
ers,templates}}}
arij@arij:~$ cd portfolio-deploy
arij@arij:~/portfolio-deploy$
```

Etape_3.Configurer les fichiers de base

3.1 ansible.cfg

```
arij@arij:~/portfolio-deploy$ cat > ansible.cfg <<'EOF'
[defaults]
inventory = ./inventory/hosts.ini
roles_path = ./roles
host_key_checking = False
retry_files_enabled = False
stdout_callback = yaml
interpreter_python = auto_silent
EOF
arij@arij:~/portfolio-deploy$
```

3.2 inventory/hosts.ini

```
  GNU nano 7.2                                    inventory/hosts.ini *
[webservers]
prod-server ansible_host=192.168.159.20 ansible_user=deploy ansible_ssh_private_key_file=~/.ssh/id_rsa
```

3.3 group_vars/all.yml

```
arij@arij:~/portfolio-deploy$ cat > group_vars/all.yml <<'EOF'
app_repo: "https://github.com/arijbelaid/deploy_Portfolio.git"
app_version: "main"
app_dir: "/opt/portfolio"
app_port: 3000
server_name: "portfolio.local"
EOF
```

Etape_4. Écrire les rôles Ansible

4.1. Rôle common

```
arij@arij:~/portfolio-deploy$ cat > roles/common/tasks/main.yml <<'EOF'
---
- name: Mettre à jour le cache apt
  apt:
    update_cache: yes
    cache_valid_time: 3600

- name: Installer les paquets de base
  apt:
    name:
      - git
      - curl
      - build-essential
      - python3-pip
    state: present

- name: Créer l'utilisateur deploy
  user:
    name: deploy
    shell: /bin/bash
    create_home: yes
    state: present
EOF
arij@arij:~/portfolio-deploy$
```

4.2 Rôle nodejs

```
arij@arij:~/portfolio-deploy$ cat > roles/nodejs/tasks/main.yml <<'EOF'
---
- name: Ajouter la clé GPG NodeSource
  apt_key:
    url: https://deb.nodesource.com/gpgkey/nodesource.gpg.key
    state: present

- name: Ajouter le dépôt NodeSource pour Node.js 20
  apt_repository:
    repo: "deb https://deb.nodesource.com/node_20.x nodistro main"
    state: present
    update_cache: yes

- name: Installer Node.js
  apt:
    name: nodejs
    state: present

- name: Installer PM2 globalement
  npm:
    name: pm2
    global: yes
    state: present
EOF
arij@arij:~/portfolio-deploy$
```

4.3 Rôle app :

```
---
- name: Cloner ou mettre à jour le dépôt de l'application
  git:
    repo: "{{ app_repo }}"
    dest: "{{ app_dir }}"
    version: "{{ app_version }}"
    force: yes
  notify: restart app

- name: Installer les dépendances npm
  command: npm ci
  args:
    chdir: "{{ app_dir }}"
  changed_when: false

- name: Construire l'application Next.js
  command: npm run build
  args:
    chdir: "{{ app_dir }}"
  changed_when: false

- name: Vérifier si l'application tourne avec PM2
  shell: pm2 list | grep -q "portfolio"
  register: pm2_status
  changed_when: false
  failed_when: false

- name: Démarrer l'application avec PM2 (si non démarrée)
  command: pm2 start npm --name "portfolio" -- start
  args:
    chdir: "{{ app_dir }}"
  when: pm2_status.rc != 0

- name: Sauvegarder la liste des processus PM2
  command: pm2 save
  args:
```

```
arij@arij:~/portfolio-deploy$ cat > roles/app/handlers/main.yml <<'EOF'
---
- name: restart app
  command: pm2 restart portfolio
  args:
    chdir: "{{ app_dir }}"
EOF
arij@arij:~/portfolio-deploy$
```

4.4 Rôle nginx

```
arij@arij:~/portfolio-deploy$ cat > roles/nginx/tasks/main.yml <<'EOF'
---
- name: Installer Nginx
  apt:
    name: nginx
    state: present

- name: Déployer la configuration du vhost
  template:
    src: portfolio.conf.j2
    dest: /etc/nginx/sites-available/portfolio.conf
  notify: reload nginx

- name: Activer le site
  file:
    src: /etc/nginx/sites-available/portfolio.conf
    dest: /etc/nginx/sites-enabled/portfolio.conf
    state: link
  notify: reload nginx

- name: Désactiver le site par défaut
  file:
    path: /etc/nginx/sites-enabled/default
    state: absent
  notify: reload nginx

- name: S'assurer que Nginx est démarré et activé
  service:
    name: nginx
    state: started
    enabled: yes
EOF
arij@arij:~/portfolio-deploy$
```

roles/nginx/handlers/main.yml

```
arij@arij:~/portfolio-deploy$ cat > roles/nginx/handlers/main.yml <<'EOF'
---
- name: reload nginx
  service:
    name: nginx
    state: reloaded
EOF
arij@arij:~/portfolio-deploy$
```

roles/nginx/templates/portfolio.conf.j2

```
arij@arij:~/portfolio-deploy$ cat > roles/nginx/templates/portfolio.conf.j2 <<'EOF'
server {
    listen 80;
    server_name {{ server_name }};

    location / {
        proxy_pass http://127.0.0.1:{{ app_port }};
        proxy_http_version 1.1;
        proxy_set_header Upgrade $http_upgrade;
        proxy_set_header Connection 'upgrade';
        proxy_set_header Host $host;
        proxy_set_header X-Real-IP $remote_addr;
        proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
        proxy_set_header X-Forwarded-Proto $scheme;
        proxy_cache_bypass $http_upgrade;
    }
}
EOF
arij@arij:~/portfolio-deploy$
```

Etape_5. Écrire le playbook principal site.yml

```
arij@arij:~/portfolio-deploy$ cat > site.yml <<'EOF'
---
- name: Déployer le portfolio Next.js
  hosts: webservers
  become: yes
  roles:
    - common
    - nodejs
    - app
    - nginx
EOF
arij@arij:~/portfolio-deploy$
```

Etape_6. Lancer Ansible

6.1. Vérification syntaxique

```
arij@arij:~/portfolio-deploy$ ansible-playbook -i inventory/hosts.ini site.yml --syntax-check

playbook: site.yml
```

6.2.Exécution en mode check (dry-run)

```
arij@arij:~/portfolio-deploy$ ansible-playbook -i inventory/hosts.ini site.yml

PLAY [Déployer le portfolio Next.js] ***********************************************************************************************

TASK [Gathering Facts] *************************************************************************************************************
ok: [prod-server]

TASK [common : Mettre à jour le cache apt] *****************************************************************************************
ok: [prod-server]

TASK [common : Installer les paquets de base] **************************************************************************************
ok: [prod-server]

TASK [common : Créer l'utilisateur deploy] *****************************************************************************************
ok: [prod-server]

TASK [nodejs : Installer Node.js depuis les dépôts Ubuntu] *************************************************************************
ok: [prod-server]

TASK [nodejs : Installer npm] ******************************************************************************************************
ok: [prod-server]

TASK [nodejs : Installer PM2 globalement] ******************************************************************************************
ok: [prod-server]

TASK [app : Cloner ou mettre à jour le dépôt de l'application] *********************************************************************
changed: [prod-server]

TASK [app : Installer les dépendances npm] *****************************************************************************************
ok: [prod-server]

TASK [app : Construire l'application Next.js] **************************************************************************************
ok: [prod-server]

TASK [app : Vérifier si l'application tourne avec PM2] *****************************************************************************
```

```
TASK [nginx : S'assurer que Nginx est démarré et activé] ***************************************************************************
ok: [prod-server]

RUNNING HANDLER [app : restart app] ************************************************************************************************
changed: [prod-server]

RUNNING HANDLER [nginx : reload nginx] *********************************************************************************************
changed: [prod-server]

PLAY RECAP *************************************************************************************************************************
prod-server                : ok=20   changed=8    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
```

6.3. Première exécution réelle :

```
arij@arij:~/portfolio-deploy$ ansible-playbook -i inventory/hosts.ini site.yml

PLAY [Déployer le portfolio Next.js] ***********************************************************************************************

TASK [Gathering Facts] *************************************************************************************************************
ok: [prod-server]

TASK [common : Mettre à jour le cache apt] *****************************************************************************************
ok: [prod-server]

TASK [common : Installer les paquets de base] **************************************************************************************
ok: [prod-server]

TASK [common : Créer l'utilisateur deploy] *****************************************************************************************
ok: [prod-server]

TASK [nodejs : Installer Node.js depuis les dépôts Ubuntu] *************************************************************************
ok: [prod-server]

TASK [nodejs : Installer npm] ******************************************************************************************************
ok: [prod-server]

TASK [nodejs : Installer PM2 globalement] ******************************************************************************************
ok: [prod-server]

TASK [app : Cloner ou mettre à jour le dépôt de l'application] *********************************************************************
ok: [prod-server]

TASK [app : Installer les dépendances npm] *****************************************************************************************
ok: [prod-server]

TASK [app : Construire l'application Next.js] **************************************************************************************
ok: [prod-server]
```

```
TASK [app : Démarrer l'application avec PM2 (si non démarrée)] *********************************************************************
skipping: [prod-server]

TASK [app : Sauvegarder la liste des processus PM2] ********************************************************************************
ok: [prod-server]

TASK [nginx : Installer Nginx] *****************************************************************************************************
ok: [prod-server]

TASK [nginx : Déployer la configuration du vhost] **********************************************************************************
ok: [prod-server]

TASK [nginx : Activer le site] *****************************************************************************************************
ok: [prod-server]

TASK [nginx : Désactiver le site par défaut] ***************************************************************************************
ok: [prod-server]

TASK [nginx : S'assurer que Nginx est démarré et activé] ***************************************************************************
ok: [prod-server]

PLAY RECAP *************************************************************************************************************************
prod-server                : ok=17   changed=0    unreachable=0    failed=0    skipped=1    rescued=0    ignored=0
```

6.4.Vérifier que le portfolio est accessible

```
arij@arij:~/portfolio-deploy$ ssh -i ~/.ssh/id_rsa deploy@192.168.159.20 "sudo pm2 status"
```

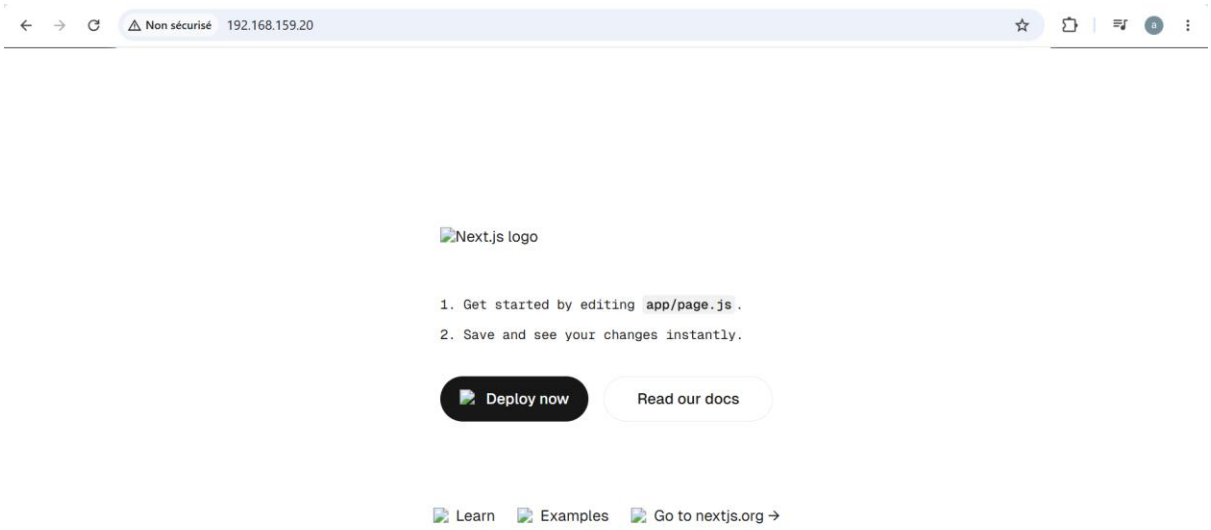
id	name	namespace	version	mode	pid	uptime	↺	status	cpu	mem	user	watching
0	portfolio	default	N/A	fork	13895	5m	1	online	0%	72.5mb	root	disabled

```
host metrics | cpu: 0.5% | ram usage: 17.1% | disk: ⇓ 0mb/s ⇑ 0.014mb/s
arij@arij:~/portfolio-deploy$
```

6.5. Vérifier que Nginx tourne

```
arij@arij:~/portfolio-deploy$ ssh -i ~/.ssh/id_rsa deploy@192.168.159.20 "sudo systemctl status nginx --no-pager | head -5"
● nginx.service - A high performance web server and a reverse proxy server
     Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: enabled)
     Active: active (running) since Fri 2026-10-16 22:59:06 UTC; 6min ago
       Docs: man:nginx(8)
    Process: 13610 ExecStartPre=/usr/sbin/nginx -t -q -g daemon on; master_process on; (code=exited, status=0/SUCCESS)
arij@arij:~/portfolio-deploy$
```

6.6. Vérifier depuis votre navigateur



Etape_7 : Lancer Semaphore UI

7.1.Lancement :

```
arij@arij:~/portfolio-deploy$ docker run -d --name semaphore \
  -p 3001:3000 \
  -e SEMAPHORE_DB_DIALECT=sqlite \
  -e SEMAPHORE_DB=/var/lib/semaphore/semaphore.sqlite \
  -e SEMAPHORE_ADMIN=admin \
  -e SEMAPHORE_ADMIN_PASSWORD=admin123 \
  -e SEMAPHORE_ADMIN_NAME=Admin \
  -e SEMAPHORE_ADMIN_EMAIL=admin@localhost \
  -e SEMAPHORE_ACCESS_KEY_ENCRYPTION=$(head -c32 /dev/urandom | base64) \
  -v semaphore_data:/var/lib/semaphore \
  semaphoreui/semaphore:latest
1e2958c6bae2c8c0e5f64b1b909b82b52fb7e8a768d40222b15b3caa04d24e42
```

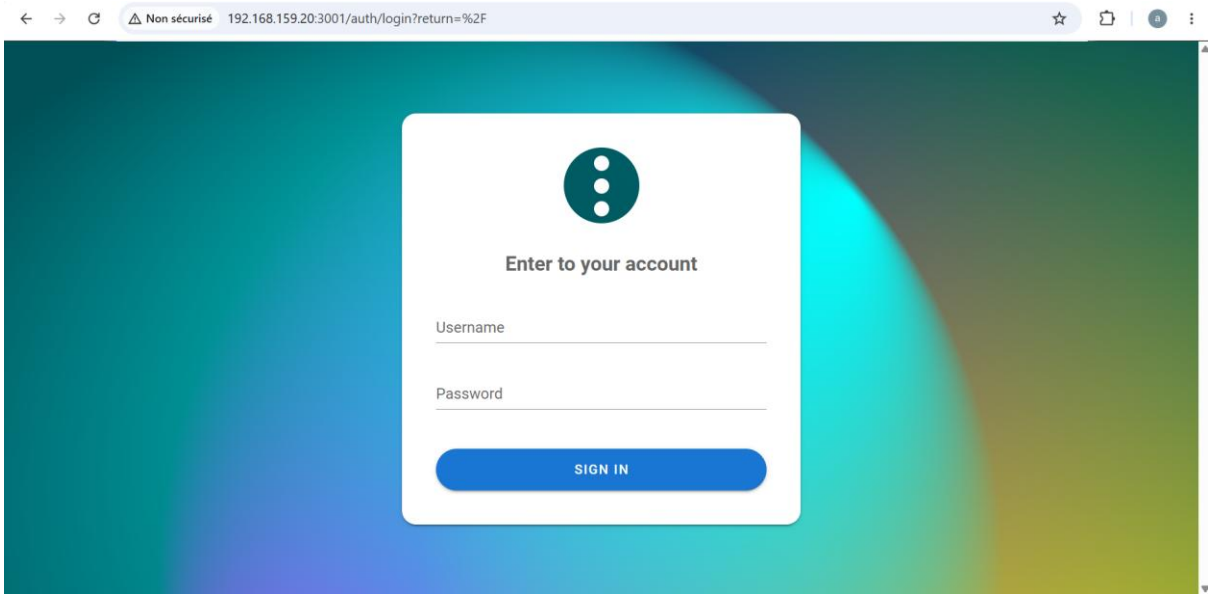
7.2.verifier l'état

```
arij@arij:~/portfolio-deploy$ docker ps | grep semaphore
1e2958c6bae2   semaphoreui/semaphore:latest   "/sbin/tini -- /usr/…"   9 seconds ago   Up 9 seconds   0.0.0.0:3001->3000/tcp, [::]:3001->3000/tc
p   semaphore
arij@arij:~/portfolio-deploy$ docker ps -a | grep semaphore
1e2958c6bae2   semaphoreui/semaphore:latest   "/sbin/tini -- /usr/…"   About a minute ago   Up About a minute   0.0.0.0:3001->3000/tcp, [::]:300
1->3000/tcp   semaphore
```

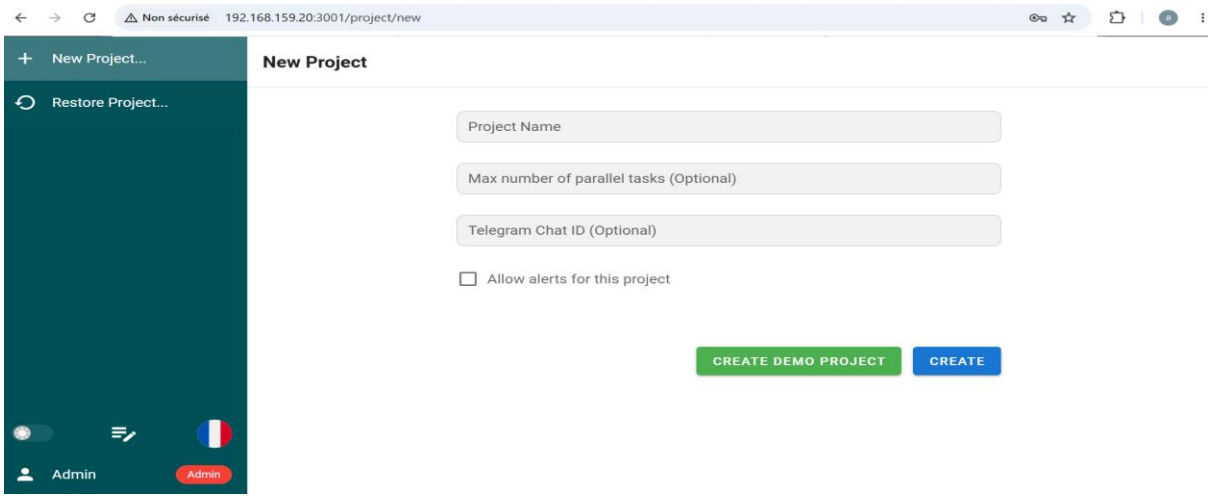
7.3.Test http :

```
arij@arij:~/portfolio-deploy$ curl -I http://localhost:3001
HTTP/1.1 200 OK
Accept-Ranges: bytes
Content-Length: 815
Content-Type: text/html; charset=utf-8
Last-Modified: Fri, 16 Oct 2026 23:17:13 GMT
Date: Fri, 16 Oct 2026 23:18:54 GMT
```

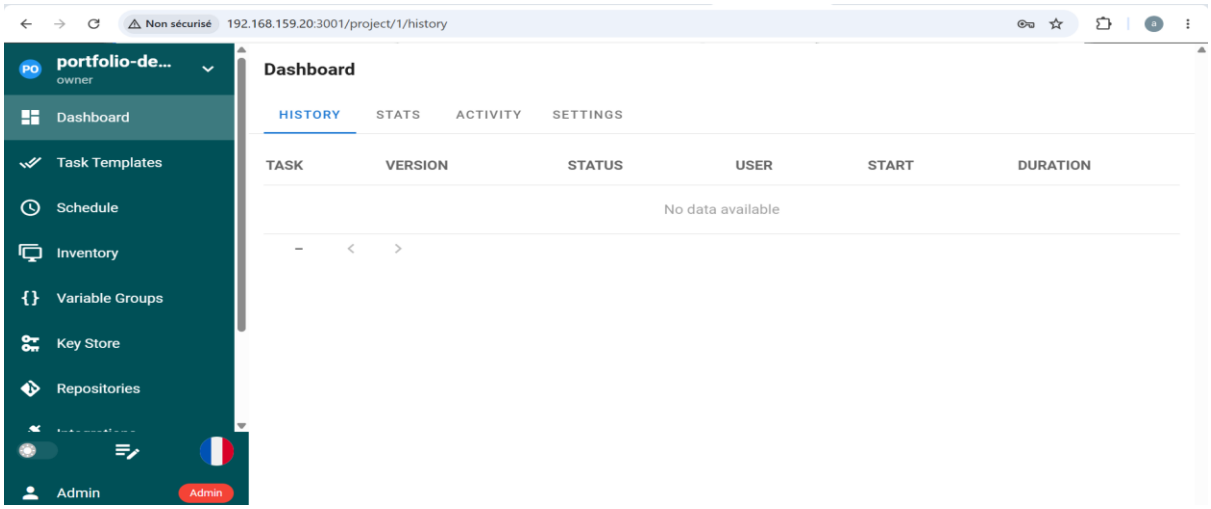
7.4.La page de login Semaphore



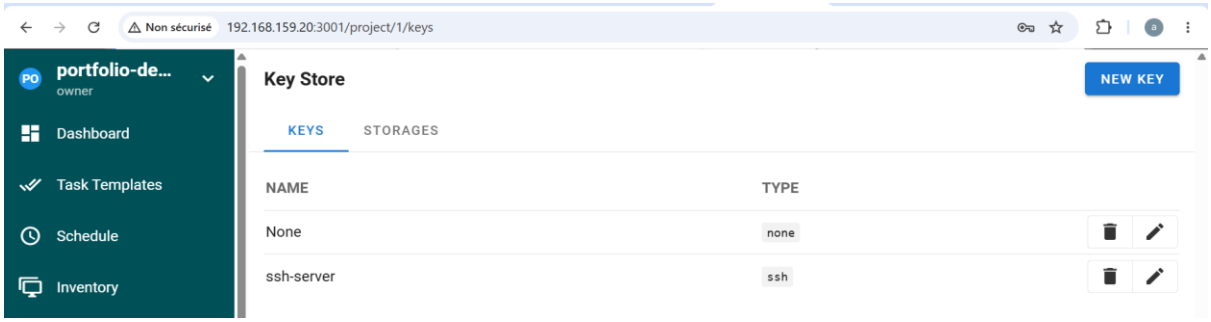
7.5.interface Semaphore :



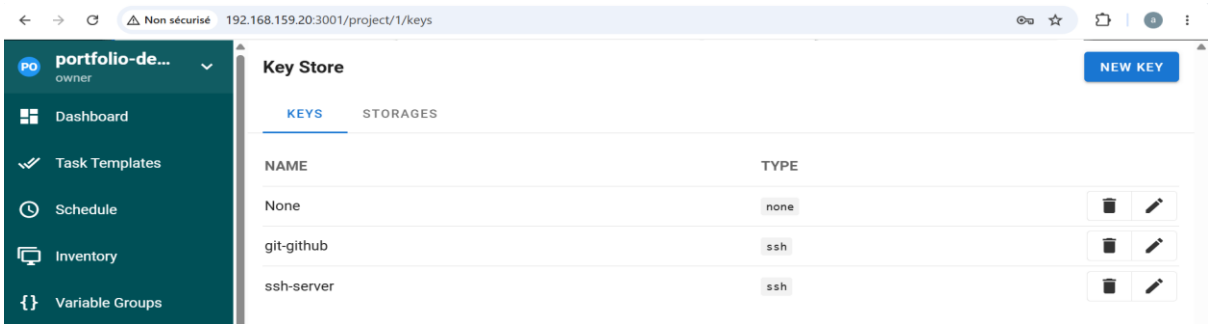
7.6. Créer le projet portfolio-deploy



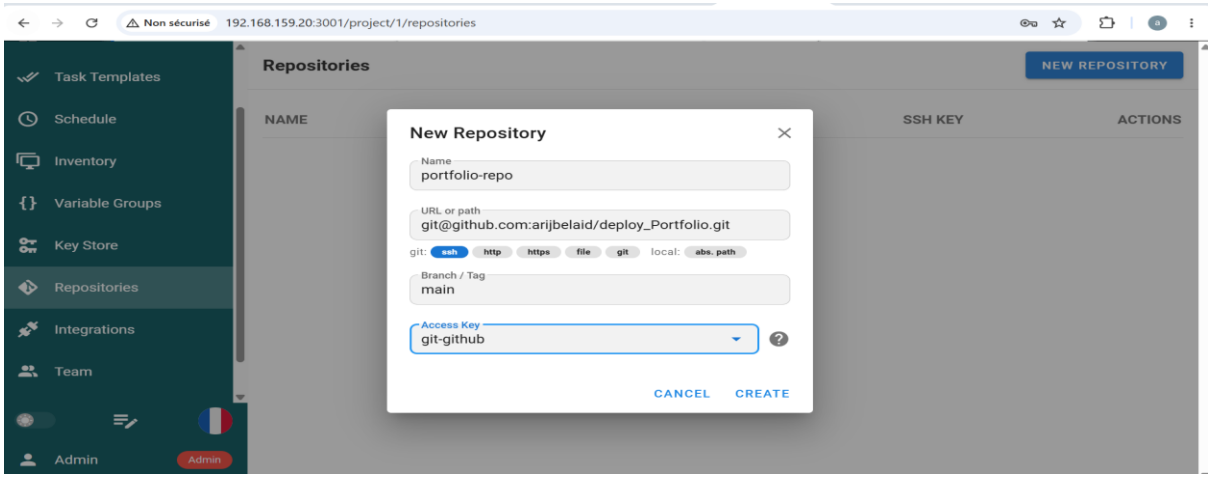
Etape_8 : Ajouter la clé SSH du serveur



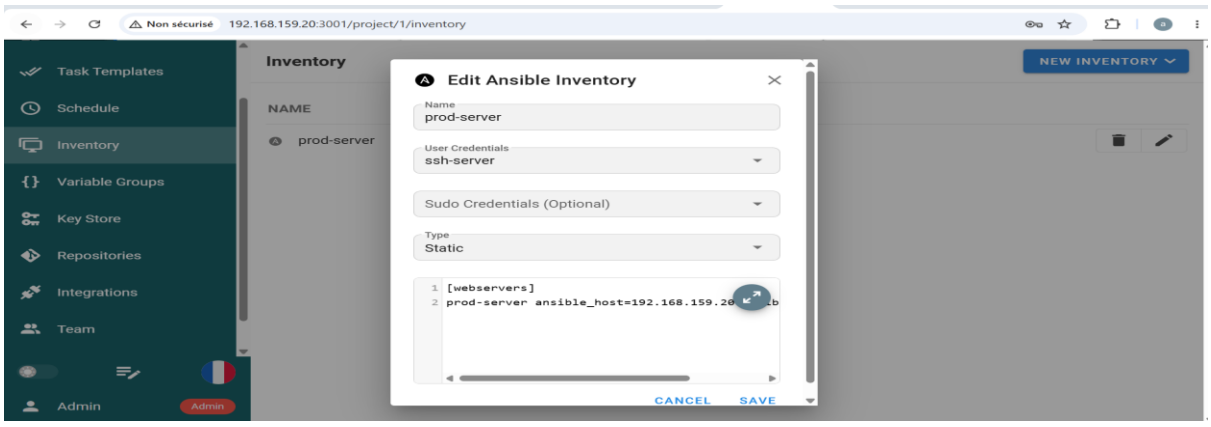
Etape_9 : Créer la clé git dans Semaphore



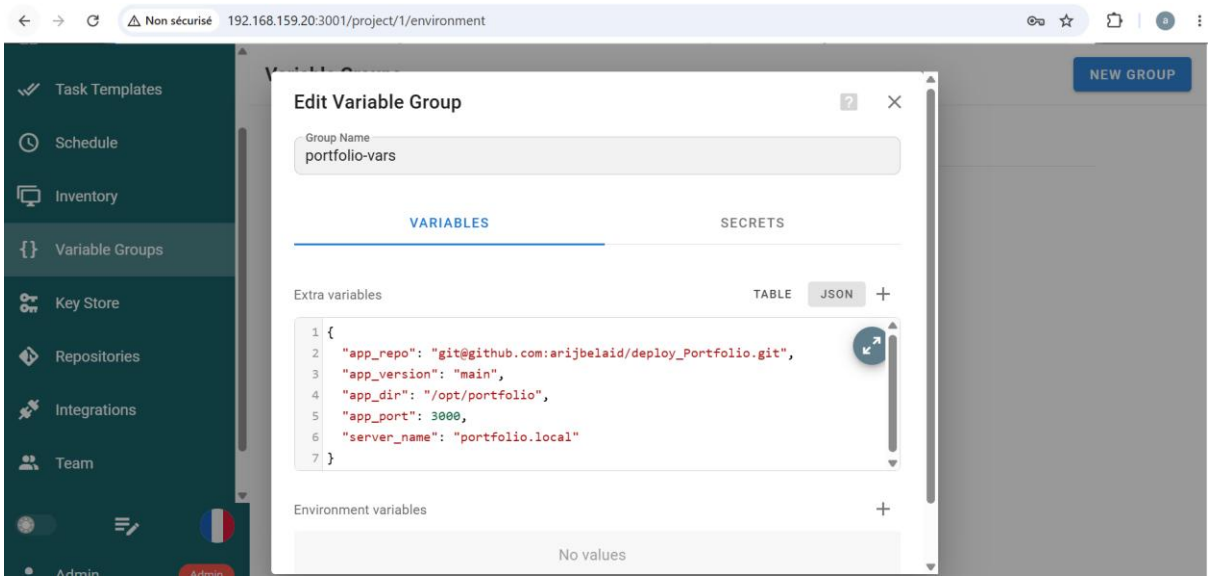
Etape_10 : Ajouter le dépôt Git



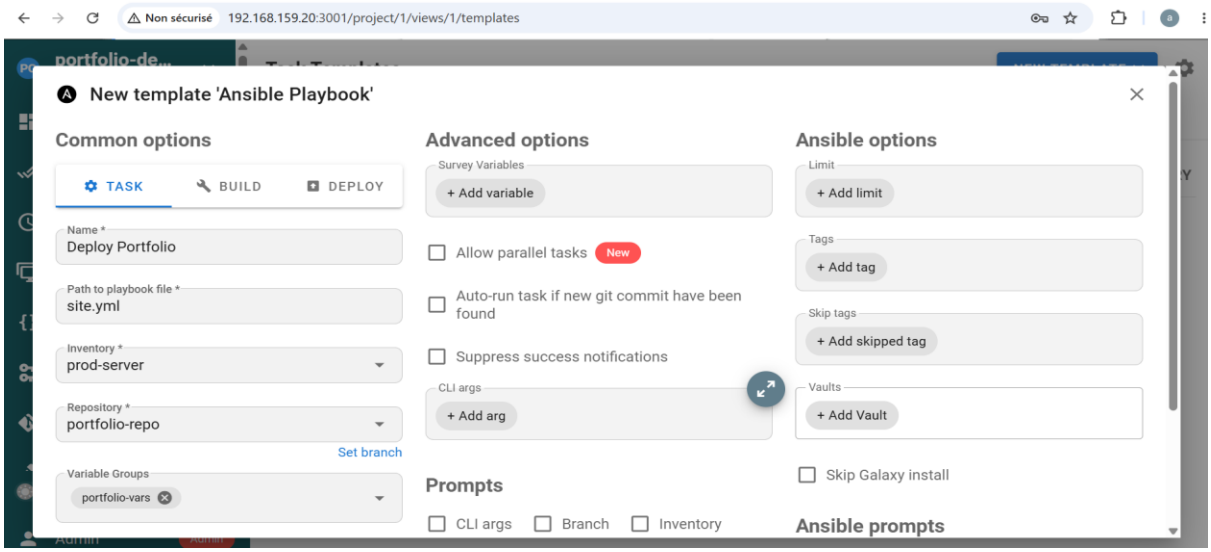
Etape_11 : Ajouter l'inventaire



Etape_12 : Ajouter le groupe de variables



Etape_13 : Créer le Task Template



Etape_14 : Lancer le déploiement

